



Inferring Individualized Color-Vision Distortions from fMRI Hue-Representation Geometry

Junil Kim¹ Albert Minkue Cho² Jungwoo Seo³ Jiook Cha^{2,5,6,*}

¹Linguistics · ²Psychology · ³Computer Science & Engineering · ⁴Brain & Cognitive Sciences · ⁵Interdisciplinary Program in AI — Seoul National University



Code & data



Repro notebooks

Structured Data for Health (SD4H) Workshop · ICML 2026

Many health conditions leave **structured — not absent — signatures** in high-dimensional measurements. We turn one into a **low-dimensional, invertible, individualized correction** — read off a person's own cortex. Testbed: color-vision deficiency (fMRI).

KEY CLAIM

In color-vision deficiency (CVD), the **continuous geometry** of cortical hue representations is distorted while **categorical recognition is preserved**.

We quantify each individual's distortion, fit a **2-parameter cortical model**, and **analytically invert** it into a per-person correction filter — exact for every displayed hue (**residual < 0.001°**).

BACKGROUND & GAP

- Many health phenotypes are **structured distortions** of a high-dimensional response space (cardiac shape, gene-expression latent shifts). CVD is a tractable, ground-truth-checkable instance.
- Such structured signatures are rarely turned into individualized corrections. The closest analogue (cochlear WHIS) simulates loss — it does not correct it.
- CVD ≈ 8% of males; cortical distortion **varies across individuals** even within one diagnostic category. Generic notch filters shift appearance but barely change discrimination.
- Gap** → an individualized, invertible correction read from each person's own cortex.

THIS WORK — THE TEMPLATE, IN 3 STEPS

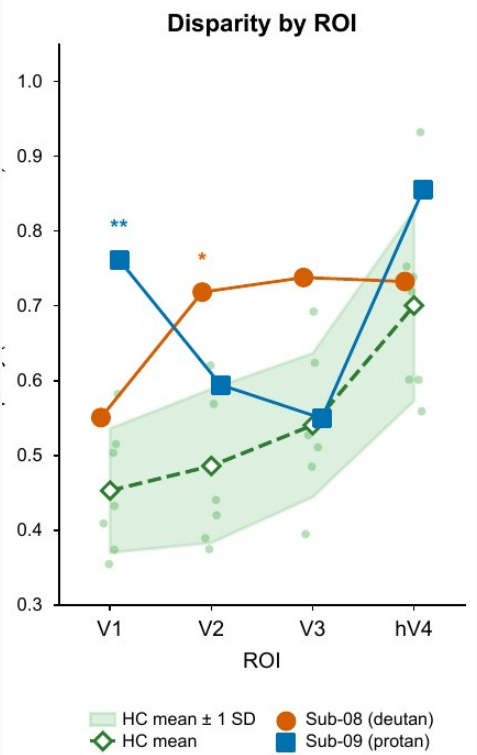
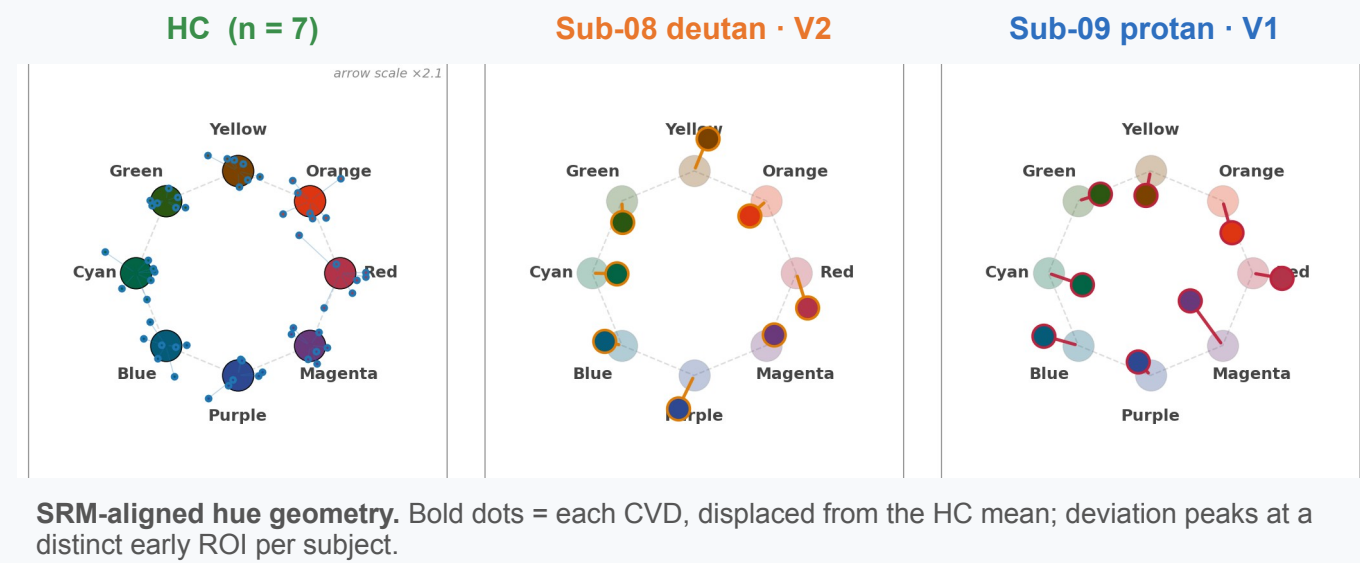
- Represent** cast the distorted neural geometry as a structured-distortion object — an **interpolation-vulnerability profile** (v) + **pairwise Δ RDM** — invisible to classification.
- Fit** a low-dimensional **2-DOF interpretable** model, fusing fragmented behavioral + neural views, selected by **held-out loss**.
- Invert** analytically invert it into a **deployable, subject-specific** correction — and test whether it works (2nd MRI).

METHODS — DATA & READOUTS

- N = 10**: 7 HC (3F, 22.7±2.5 y) + 3 CVD (2 deutan, 1 protan), Ishihara-confirmed. Small sample → **single-case statistics** (Crawford–Howell).
- Stimuli**: 8 isoluminant hues (CIE $L^*a^*b^*$, $L^*=75$, chroma 40), RSVP, 6 runs, Siemens 3T → fMRIprep → V1/V2/V3/hV4 (Wang 50%); FIR-GLM, Procrustes-aligned. JND outside scanner.
- One forward-encoding model** ($F = 6$ channels), two hold-outs: **LORO** (hold a run → classification) vs **LOCO** (hold a hue → interpolation).
- Signatures**: vulnerability $v \in [0, 1]^8$ at hV4 + Δ RDM = RDM(CVD) – mean RDM(HC), 28 pairs.

RESULTS

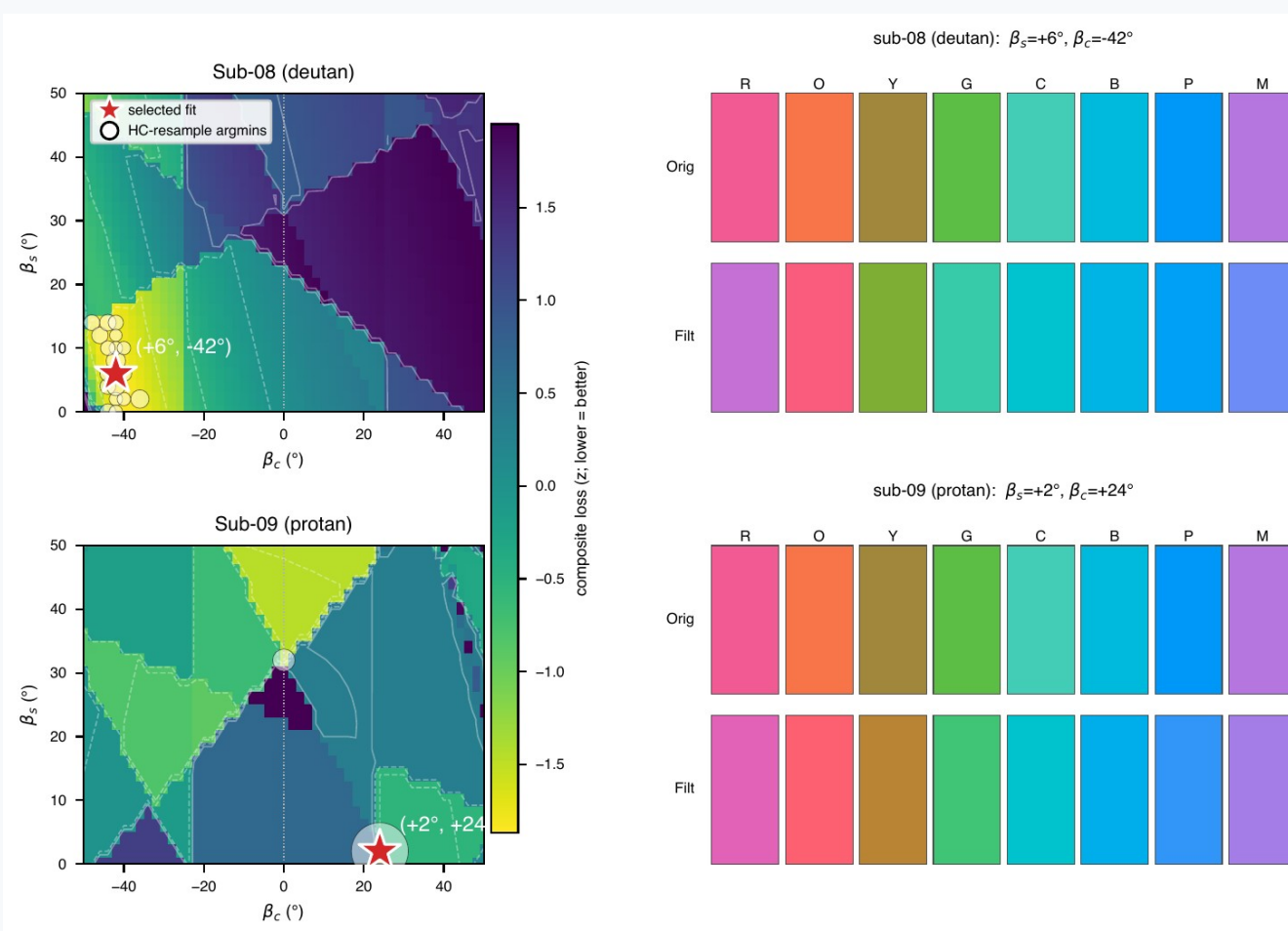
A. A structured cortical distortion



SRM disparity by ROI — each CVD exceeds the HC ±1 SD band at one ROI.

- Discrimination preserved**. Cross-run (LORO) classification: no HC–CVD difference (pooled $p=0.668$).
- Interpolation impaired**. Only hV4 interpolates above chance in HC (adjacent acc 0.47, $p=0.044$); both CVD below — deficit on S-cone hues.
- Distinct ROI per subject**. Disparity elevated at Sub-08 **V2** ($p=0.040$), Sub-09 **V1** ($p=0.007$) — Crawford–Howell.

B. An individualized, invertible filter



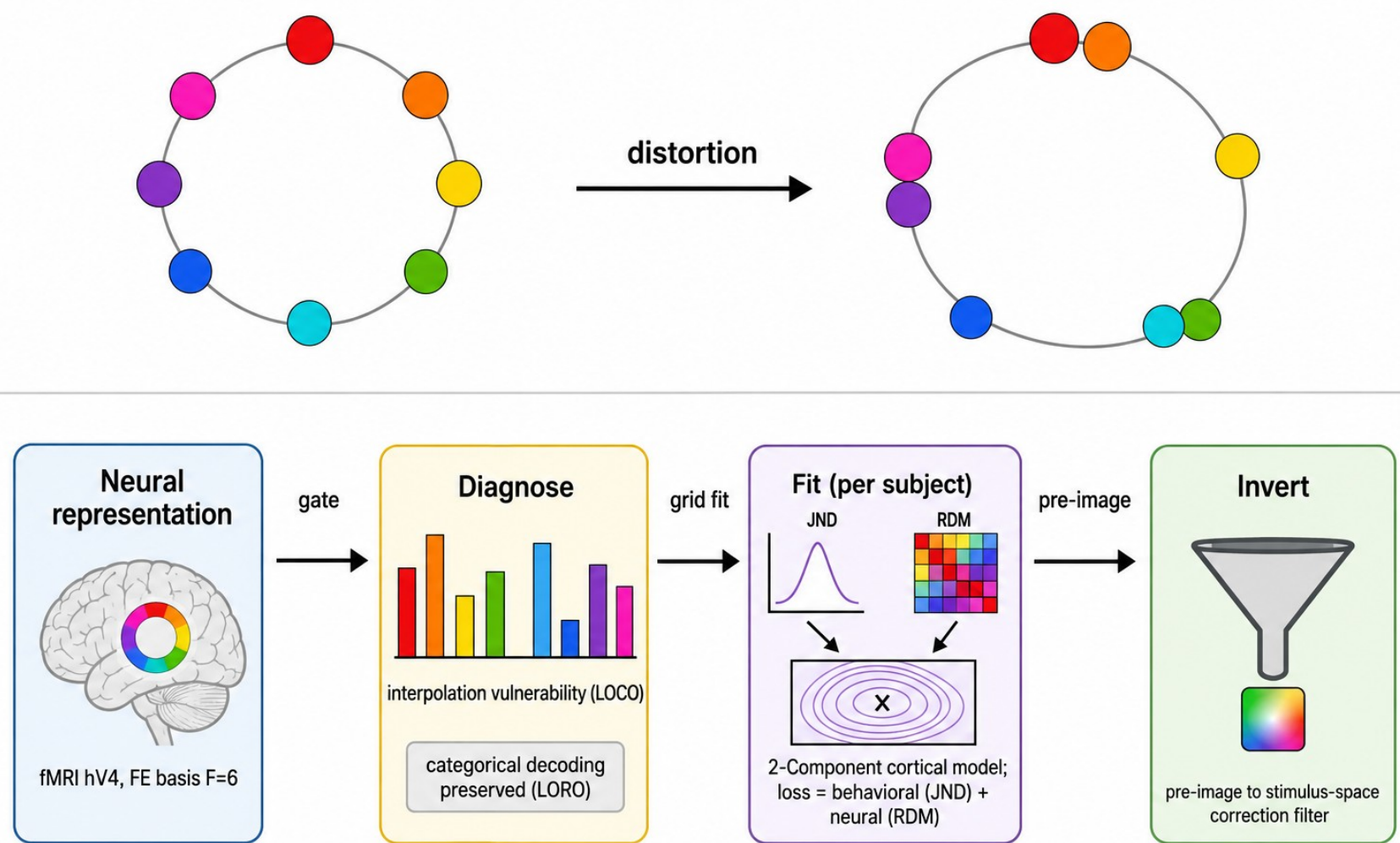
Per-subject 2-Component fit (left: loss landscape over (β_s, β_c) ; right: inverted filter, original vs corrected hues).

Table 1 — selected 2-Component fit (hV4)

Subject	(β_s, β_c)	Test loss	$ \delta\theta $
Sub-08 deutan	$(+6^\circ, -42^\circ)$	–2.36	26.3°
Sub-09 protan	$(+2^\circ, +24^\circ)$	–1.54	16.2°

- Structure beats cone-shift**. R+C overcompensates (gain pinned at ceiling) and is **non-invertible at protan severity** (4/8 hues lack a pre-image). 2-Component fits the grid interior for both.
- Neural term carries what behavior cannot**. For Sub-09, behavioral-only beats the null in 3/7 folds; adding neural Δ RDM → 7/7 + a stable confusion-axis rotation.
- Exact inversion**. A pre-image for all 8 hues (residual < 0.001°) → a directly implementable 8-point correction LUT; filters differ by subtype (β_c –42° deutan vs +24° protan).

FRAMEWORK — DIAGNOSE → FIT → INVERT



Top: CVD distorts the continuous hue circle — neighbors collapse or expand — though classification stays intact. **Bottom**: per-subject pipeline — diagnose, fit a 2-parameter cortical model to behavioral + neural loss, invert to a stimulus-space filter.

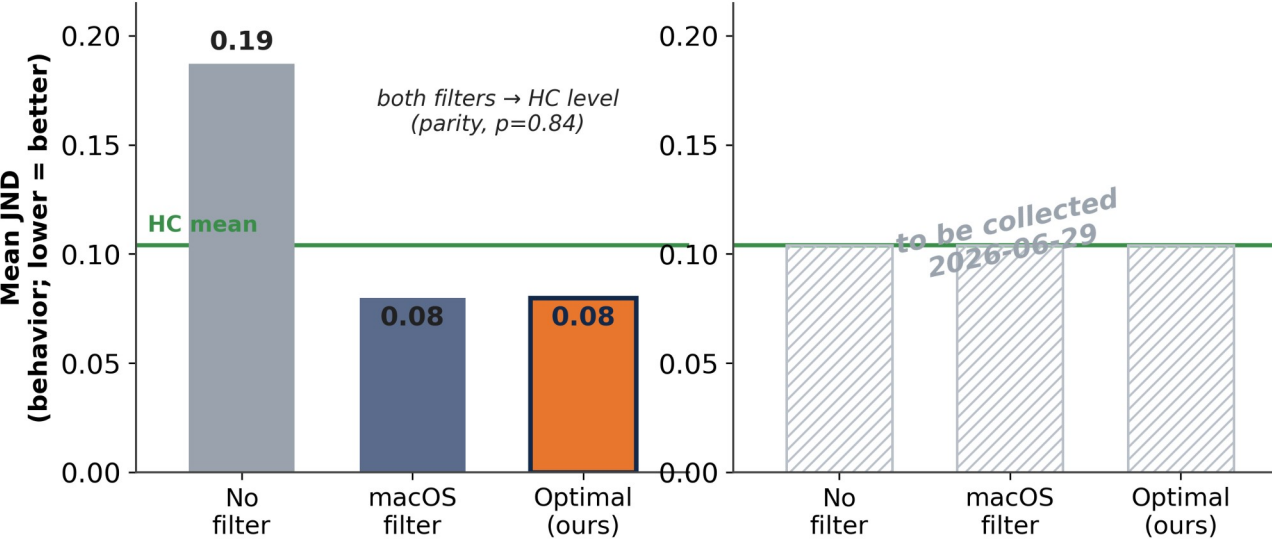
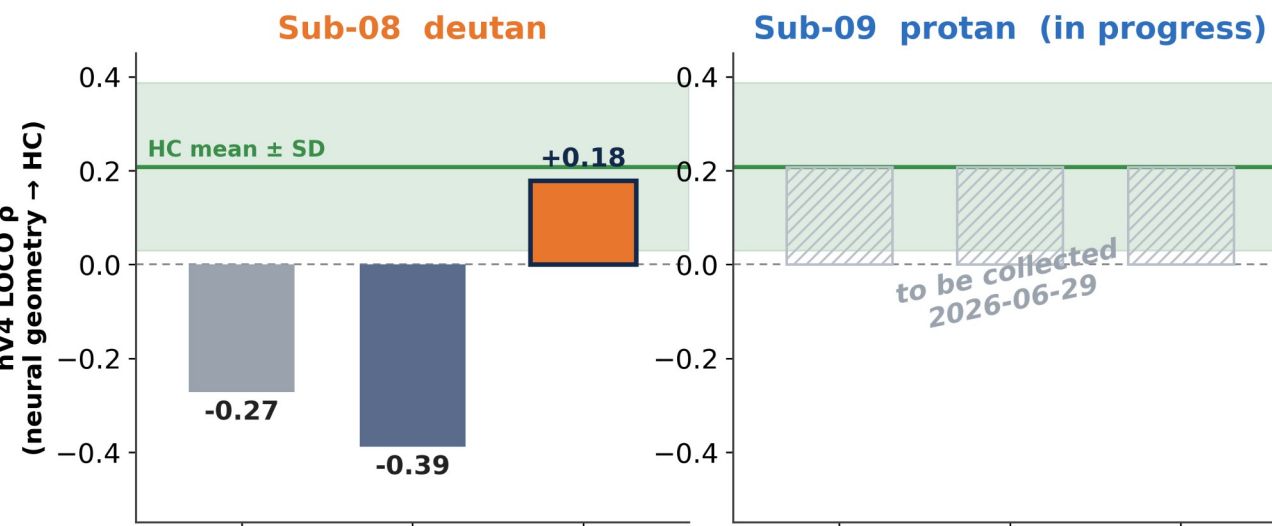
MODELS & FITTING

Mechanism	DOF	Form
Retinal cone-shift (Machado)	1	$\Delta\lambda$
Retinal + cortical (R+C)	1	$\delta\theta = (2-g)\delta\theta_{\text{Machado}}$
Cortical 2-Component	2	$\theta' = \theta + \beta_s \cos(\theta-90^\circ) + \beta_c \cos(\theta-\theta_{\text{conf}})$

- β_s = S-cone axis, β_c = confusion axis ($\theta_{\text{conf}} 16^\circ$ protan / 150° deutan).
- Loss = **behavioral JND-ratio rank** + **neural Δ RDM cosine**, selected by **held-out generalization** (7-fold leave-one-HC-out) — not in-sample significance.
- Invert**: the 2-Component map is bijective → per-hue pre-image $\hat{\theta}_k$ by root-finding; correction $\delta\theta_k = \hat{\theta}_k - \theta_k$.

C. Does the filter work? (2nd MRI)

Behavioral parity, but only the model-derived filter repairs the cortex



CVD subject views the 8 hues through each filter in-scanner (4 runs/condition); **Optimal** = personalized inverse filter, **macOS** = deployed generic filter.

- Behavioral parity**. The personalized filter restores discrimination to HC level (mean JND 0.19→0.08; HC 0.10), **on par with the deployed macOS filter** (Wilcoxon $p=0.84$).
- Neural superiority**. Yet **only the model-derived filter** restores hV4 interpolation geometry toward HC ($p+0.18 \approx$ HC 0.21); the macOS filter pushes it further away (–0.39). $\Delta = 3.2$ HC-SD, run-separated.
- Scope**. Descriptive proof-of-concept (Sub-08; $n=4$ /condition, no permutation; macOS vs PsychoPy rendering differs). **Sub-09 added 2026-06-29 → N=2**.

CONCLUSION & NEXT STEP

- A worked template**: quantify a structured fMRI distortion, fit a parsimonious interpretable model, **analytically invert** it into a deployable per-person filter.
- Validated (proof-of-concept)**: the personalized filter matches a deployed generic filter behaviorally, and **uniquely repairs the cortical hue geometry** (hV4). Sub-09 in progress.
- For SD4H**: mapping structured signatures onto low-dimensional, invertible features turns descriptive phenotypes into individualized intervention targets.

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Correspondence: connectome@snu.ac.kr · github.com/Transconnectome/colorBlind_analysis